01 Methods of Differentiation

1. MEANING OF DERIVATIVE OR $\frac{dy}{dx}$

Let $y = f(x)$; $y + \Delta y = f(x + \Delta x)$

$\therefore \frac{\Delta y}{\Delta x} = \frac{f(x+\Delta x)-f(x)}{\Delta x}$ (average rate of change of function)

$\therefore \frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x+\Delta x)-f(x)}{\Delta x}$ ...(i)

Physical Meaning:

It denotes the instantaneous rate of change of function

Geometrical Meaning:

It denotes the slope of tangent at any point on curve $y = f(x)$.

Note:

(i) If $y = f(x)$ then the symbols $\frac{dy}{dx} = Dy = f'(x) = y_1$ or y' have the same meaning.

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(ii) $f''(x) = \frac{d}{dx}\left(f'(x)\right) = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = y_2 = D^2y$

(iii) If $y = f(x)$ then $f'(a)$ means value of $f'(x)$ at $x = a$

(iv) However, a dot, the symbol denotes the time derivative, e.g. $\dot{S} = \frac{dS}{dt}$; $\dot{\theta} = \frac{d\theta}{dt}$ etc.

Derivative by First Principle:

Finding the value of the limit given by (i) in respect of **variety** of functions is called finding the derivative by first principle/by delta method/by ab-initio method/by fundamental definition of calculus.

Illustration 1

Find derivative by using 1st principle of $f(x) = x^n$; $n \neq 1$

Solution:

$$f'(x) = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = \lim_{h\to 0} \frac{(x+h)^n-(x)^n}{h}$$

$$\lim_{h\to 0} \frac{x^n}{h}\left[\left(1+\frac{h}{x}\right)^n - 1\right] = \lim_{h\to 0} \frac{x^n}{h}\left[1+\frac{nh}{x}+\frac{n(n-1)h^2}{x^2 2!}.....-1\right]$$

$$\lim_{h\to 0} x^n\left[\frac{n}{x}+h(.....\text{terms})\right]$$

$$= n\,x^{n-1}$$

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Illustration 2

Find derivative by using 1st principle of $f(x) = a^x, a > 0, a \neq 1$

Solution:

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{a^{x+h}-a^x}{h}$$

$$\Rightarrow \lim_{h\to 0} a^x\left(\frac{a^h-1}{h}\right) = a^x \ell na$$

Illustration 3

Find derivative by using 1st principle of $f(x) = \sin x$.

Solution:

$$f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{\sin(x+h)-\sin x}{h} = \lim_{h\to 0}\frac{2\cos\left(x+\frac{h}{2}\right)\sin\left(\frac{h}{2}\right)}{h}$$

$$= \lim_{h\to 0}\frac{\cos\left(x+\frac{h}{2}\right)\cdot\sin\left(\frac{h}{2}\right)}{h/2}$$

$$= \cos x \,.\, 1 = \cos x$$

2. DERIVATIVE OF STANDARD FUNCTIONS:

$f(x)$	$f'(x)$	$f(x)$	$f'(x)$
x^n	nx^{n-1}	$\sin x$	$\cos x$
e^x	e^x	$\cos x$	$-\sin x$
a^x	$a^x \ell na, a > 0$	$\tan x$	$\sec^2 x$
ℓnx / $\log_e x$	$\frac{1}{x}$	$\sec x$	$\sec x \tan x$
$\log_a x$	$\left(\frac{1}{x}\right)\log_a e, a > 0, a \neq 1$	$\operatorname{cosec} x$	$-\operatorname{cosec} x . \cot x$
Constant	0	$\cot x$	$-\operatorname{cosec}^2 x$

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Note:

The derivative values of some all **co**-trigonometric functions begin with a – *ve* sign.

Derivative of Inverse Trigonometric Function:

$f(x)$	$f'(x)$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$
$\cos^{-1} x$	$-\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$
$\tan^{-1} x$	$\frac{1}{1+x^2}, x \in R$
$\cot^{-1} x$	$-\frac{1}{1+x^2}, x \in R$
$\sec^{-1} x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}}, \lvert x\rvert > 1$
$\operatorname{cosec}^{-1} x$	$-\frac{1}{\lvert x\rvert\sqrt{x^2-1}}, \lvert x\rvert > 1$

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Illustration 4

If $D(y)$ denotes, derivative of y w. r. t. x then find:

(i) $D(e^{\pi})$, $D(x^{e})$

(ii) $D\left(\frac{1}{\sin x}\right)$

(iii) $D\left(\frac{\sin 2x}{1+\cos 2x}\right)$

(iv) $D(\tan(\tan^{-1} x))$

(v) $y = \left(\frac{x^a}{x^b}\right)^{a+b}\left(\frac{x^b}{x^c}\right)^{b+c}\left(\frac{x^c}{x^a}\right)^{c+a}$, find $\frac{dy}{dx}$

Solution:

(i) $D(e^{\pi}) = 0, \quad D(x^{e}) = e.x^{e-1}$

(ii) $D\left(\frac{1}{\sin x}\right) = D(\text{cosec}x) = -\text{cosec}x\,\cot x$

(iii) $D\left(\frac{\sin 2x}{1+\cos 2x}\right) = D\left(\frac{2\sin x\cos x}{1+2\cos^2 x-1}\right) = D(\tan x) = \sec^2 x$

(iv) $D(\tan(\tan^{-1} x)) = D(x) = 1$

(v) $y = \left(\frac{x^a}{x^b}\right)^{a+b}\left(\frac{x^b}{x^c}\right)^{b+c}\left(\frac{x^c}{x^a}\right)^{c+a}$

$\Rightarrow y = (x^{a-b})^{a+b}(x^{b-c})^{b+c}(x^{c-a})^{c+a}$

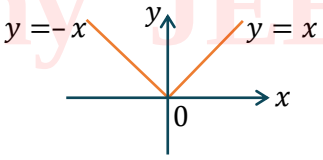
$\Rightarrow y = x^{a^2-b^2+b^2-c^2+c^2-a^2}$

$\Rightarrow y = x^0 = 1 \quad \therefore \frac{dy}{dx} = 0$

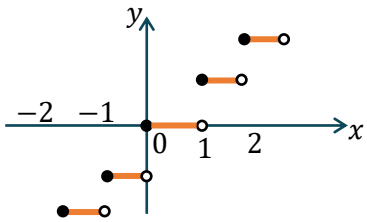
Note:

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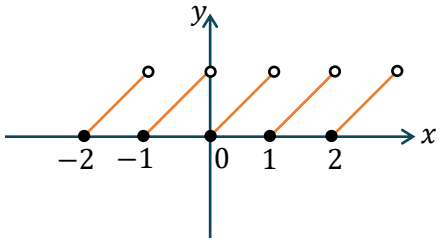
(i) $\frac{d}{dx}(|x|) = \begin{cases} 1, x > 0 \\ -1, x < 0 \\ \text{not differentiable at } x = 0 \end{cases}$



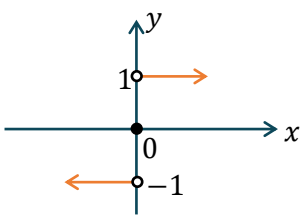
(ii) $\frac{d}{dx}([x]) = 0$, $x \notin I$ and non-differentiable at $x \in I$



(iii) $\frac{d}{dx}(\{x\}) = 1, x \notin I$ and non-differentiable at $x \in I$



(iv) $\frac{d}{dx}(sgn\,x) = 0, x \neq 0$ and non-differentiable at $x = 0$



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3. SUPPLEMENTARY THEOREMS/RESULTS:

(a) Sum/Difference Rule:

$D(f(x) \pm g(x)) = f'(x) \pm g'(x)$

Illustration 5

If $y = e^x + 3\,\ell n\,x - 4\tan^{-1}x + \sin 3x + 4\sin^3 x$, then $\frac{dy}{dx}$ is ....

Solution:

$y = e^x + 3\,\ell n\,x - 4\tan^{-1}x + \sin 3x + 4\sin^3 x$

$\Rightarrow y = e^x + 3\,\ell n\,x - 4\tan^{-1}x + 3\sin x - 4\sin^3 x + 4\sin^3 x$

$\Rightarrow y = e^x + 3\,\ell n\,x - 4\tan^{-1}x + 3\sin x$

$\Rightarrow \frac{dy}{dx} = e^x + \frac{3}{x} - \frac{4}{(1+x^2)} + 3\cos x$

(b) Product Rule:

$D(f(x).g(x)) = f(x).g'(x) + g(x).f'(x)$

Note:

If 3 functions are involved then remember

$D(f(x).g(x).h(x)) = f(x).g(x).h'(x) + g(x).h(x).f'(x) + h(x).f(x).g'(x)$

This result can be generalised to the product of n functions.

Note:

Differentiate one function at a time & keep remaining as it is and complete the process for all functions one by one.

$\frac{d}{dx}(kf(x)) = k\cdot\frac{d}{dx}f(x)$, (where k is any constant) = $\boldsymbol{kf'(x)}$

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Illustration 6

Find derivative w.r.t. x :

(i) $y = e^x + 3\,\ell n\,x - 4\sin x$

(ii) $y = x\sin^{-1}x$

(iii) $y = (x^2 e^x\,\ell n\,x)$

Solution:

(i) $y = e^x + 3\,\ell n\,x - 4\sin x \Rightarrow \frac{dy}{dx} = e^x + \frac{3}{x} - 4\cos x$

(ii) $y = x\sin^{-1}x \Rightarrow \frac{dy}{dx} = 1\cdot\sin^{-1}x + x\bullet\alpha\frac{1}{\sqrt{1-x^2}} = \sin^{-1}x + \frac{x}{\sqrt{1-x^2}}$

(iii) $y = (x^2 e^x \ell n x) \Rightarrow \frac{dy}{dx} = 2x\cdot(e^x\ell n x) + x^2(e^x)\ell n x + x^2 e^x.\frac{1}{x}$

$\Rightarrow \frac{dy}{dx} = 2xe^x\ell n x + x^2 e^x\ell n x + xe^x$

Illustration 7

If $f(x) = (1+x)(3+x^2)^{1/2}(9+x^3)^{1/3}$, then $f'(-1)$ is equal to –

(A) 0 (B) $2\sqrt{2}$ (C) 4 (D) 6

Ans. C

Solution:

$f'(x)|_{x=-1} = (x^2+3)^{1/2}(9+x^3)^{1/3} = 2.2 = 4$

($\because$ value of the other two expression in $f'(x)$ vanishes)

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(c) Quotient Rule : $y = \frac{f(x)}{g(x)}$ (Prove by First Principle)

$D\left(\frac{f(x)}{g(x)}\right) = \frac{g(x).f'(x)-f(x).g'(x)}{g^2(x)}$, to be remembered as $D\left(\frac{N^r}{D^r}\right) = \frac{D^r\frac{d}{dx}(N^r)-N^r\frac{d}{dx}(D^r)}{(D^r)^2}$

Proof:

$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}$

L.H.S. = $d\left(f(x).\frac{1}{g(x)}\right)$

$\Rightarrow \frac{d(f(x))}{dx}\cdot\frac{1}{g(x)} + f(x)\cdot\frac{d}{dx}\left(\frac{1}{g(x)}\right)$

$\Rightarrow f'(x)\cdot\frac{1}{g(x)} + f(x)\frac{(-1)}{(g(x))^2}\cdot g'(x)$ = R.H.S.

Similarly, $\frac{d\left(\frac{f(x)h(x)}{g(x)}\right)}{dx} = \frac{[f(x)h'(x)+f'(x)h(x)]g(x)-f(x)h(x)\cdot g'(x)}{(g(x))^2}$

Note:

While applying quotient rule, think twice if your function could be simplified prior to differentiation.

Consider the example

Illustration 8

$y = \frac{x^3+x^2+1}{x^2+1}$, then find $\frac{dy}{dx}$

Solution:

M-1 $= \frac{(3x^2+2x+0)(x^2+1)-(x^3+x^2+1)(2x+0)}{(x^2+1)^2}$

M-2 $f(x) = \frac{x^3}{x^2+1} + \frac{x^2+1}{x^2+1} \Rightarrow \frac{x^3}{x^2+1} + 1$ (Now apply quotient rule)

Illustration 9

$y = \frac{\sin^{-1}x-\cos^{-1}x}{\tan^{-1}x+\cot^{-1}x}$, then find $\frac{dy}{dx}$

Solution:

$y = \frac{\left(\sin^{-1}x-\cos^{-1}x\right)}{\frac{\pi}{2}} = \frac{2}{\pi}(\sin^{-1}x-\cos^{-1}x)$

$\frac{dy}{dx} = \frac{2}{\pi}\left[\frac{1}{\sqrt{1-x^2}} - \left(\frac{-1}{\sqrt{1-x^2}}\right)\right] = \frac{4}{\pi\sqrt{1-x^2}}$

General Note:

If $f'(x)$ is not defined on $x = c$ then it is wrong to conclude that $f(x)$ is not derivable at $x = c$. In such cases we find, LHD at $x = c$ and RHD at $x = c$

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Illustration 10

$f(x) = x^{\frac{1}{3}} \sin x$ at $x = 0$

Solution:

$f'(x) = x^{1/3} \cos x + \frac{1}{3} x^{-2/3} . \sin x = x^{1/3} \cos x + \frac{\sin x}{3x^{2/3}}$

$f'(0)$ seems to be non-defined but $f'(0^+)$ and $f'(0^-)$ is 0.

$f'(0^+) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$

$= \lim_{h\to 0} \frac{h^{\frac{1}{3}} \sin h}{h}$ $(\because f(0) = 0)$

$= 0$

$f'(0^-) = \lim_{h\to 0} \frac{f(0-h) - f(0)}{-h}$

$= \lim_{h\to 0} \frac{(-h)^{\frac{1}{3}} \sin(-h)}{-h} = 0$

Chain Rule:

Proof:

If $y = f(u)$ and $u = g(x)$

then $\frac{dy}{dx} = \frac{dy}{du} . \frac{du}{dx} = f'(u) \cdot g'(x) = f'(g(x)) \cdot g'(x)$.

It can be extended to any number of chains.

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Illustration 11

$\frac{d}{dx} \tan(\log x)$

Solution:

$\sec^2(\log x) \cdot \frac{1}{x}$

Illustration 12

$\frac{d}{dx} (\sin x^2)$

Solution:

$\cos x^2 . 2x$

Illustration 13

$\frac{d}{dx} \log \sin x^2$

Solution:

$= \frac{1}{\sin x^2} . \cos x^2 . 2x$

Illustration 14

$\frac{d}{dx} [\tan^{-1}(\log \sin x^2)]$

Solution:

$\frac{1}{1+(\log \sin x^2)^2} \cdot \frac{1}{\sin x^2} \cdot \cos x^2 \cdot 2x$

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4. LOGARITHMIC DIFFERENTIATION:

To find the derivative of:

(i) a function which is the product or quotient of a number of functions

(ii) a function of the form $[f(x)]^{g(x)}$ where f & g are both derivable functions, it will be found convenient to take the logarithm of the function first & then differentiate

OR

express $y = (f(x))^{g(x)} = e^{g(x)\ell n(f(x))}$ and then differentiate.

Illustration 15

If $y = x^x$, then find $\frac{dy}{dx}$

Solution:

M-1 $\log y = \log x^x$

$\Rightarrow \log y = x \log x$

$\Rightarrow \left[\frac{1}{y}\cdot\frac{dy}{dx}\right] = \left[1.\log x + x\frac{1}{x}\right]$

$\Rightarrow \frac{dy}{dx} = y\left[\log x + 1\right]$

$\Rightarrow \frac{dy}{dx} = x^x\left[\log x + 1\right]$

M-2 $y = e^{x\ell nx}$

$\Rightarrow \frac{dy}{dx} = e^{x\ell nx}\left(\frac{d}{dx}x\ell nx\right)$

$\Rightarrow x^x\left[x\cdot\frac{1}{x} + \ell nx\right] = x^x[1 + \ell nx]$

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Illustration 16

If $y = (\sin x)^{\cos x}$, then find $\frac{dy}{dx}$

Solution:

M-1 $\log y = \log(\sin x)^{\cos x}$

$\Rightarrow \log y = \cos x \,.\log(\sin x)$

$\Rightarrow \frac{1}{y}.\frac{dy}{dx} = \left[-\sin x \,.\log(\sin x) + \frac{\cos x.1.\cos x}{\sin x}\right]$

$\Rightarrow \frac{dy}{dx} = y\left[-\sin x.\log(\sin x) + \frac{\cos^2 x}{\sin x}\right]$

M-2 $y = e^{\cos x\,\ell n\sin x}$

$\Rightarrow \frac{dy}{dx} = e^{\cos x\ell n\sin x}\left(\frac{\cos x\cdot\cos x}{\sin x} + (-\sin x)\ell n\sin x\right)$

$\Rightarrow y\left(\frac{\cos^2 x}{\sin x} - \sin x\cdot\ell n\sin x\right)$

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Illustration 17

Find equation of tangent at $x = 1$ of curve $y = x^x + x^e$

Solution:

$\frac{dy}{dx} = x^x[\ell nx + 1] + e \cdot x^{e-1}$

$\left.\frac{dy}{dx}\right|_{x=1}$ = Slope at $x = 1$

$= (1)^1\ [\ell n1 + 1] + e.1^{e-1} = 1(0 + 1) + e.1 = 1 + e$

Point:

$x = 1$

$\Rightarrow y = 1 + 1 = 2$

Equation of tangent:

$y - 2 = (1 + e)\ (x - 1)$

$\Rightarrow y - 2 = x - 1 + ex - e$

$\Rightarrow y = x - 1 + 2 + ex - e$

$\Rightarrow y = x\ (1 + e) + (1 - e)$

Illustration 18

If $f(x) = (x + 1)(x + 2)(x + 3)\ldots(x + n)$ then $f'(0)$ is –

(A) $n!$ (B) $\frac{n(n+1)}{2}$ (C) $(n!)(\ell nn!)$ (D) $n!\left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \ldots + \frac{1}{n}\right)$

Ans. (D)

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Solution:

$f(x) = (x + 1)(x + 2)(x + 3)\ldots.(x + n)$

$\ell n(f(x)) = \ell n(x + 1) + \ell n(x + 2) + \ldots.. + \ell n(x + n)$

$\frac{f'(x)}{f(x)} = \frac{1}{x+1} + \frac{1}{x+2} + \ldots.. + \frac{1}{x+n}$

$\therefore f'(0) = (n)!\left(1 + \frac{1}{2} + \frac{1}{3} + \ldots\ldots\ldots + \frac{1}{n}\right)$

5. DERIVATIVE OF $f(x)$ w. r. t. $g(x)$:

If $y = f(x)$ and $z = g(x)$, then derivative of $f(x)$ w. r. t. $g(x)$ is given by

$\frac{dy}{dz} = \frac{dy}{dx} \cdot \frac{dx}{dz} = \frac{f'(x)}{g'(x)}$

$\therefore$ Differential coefficient of $f(x)$ w.r.t. $g(x) = \frac{\text{derivative of } f(x) \text{ w.r.t. } x}{\text{derivative of } f(x) \text{ w.r.t. } x} = \frac{f\prime(x)}{g\prime(x)}$

Illustration 19

Differentiate $\sin x$ w. r. t. x^3

Solution:

$\frac{d(\sin x)}{d(x^3)} = \frac{\cos x}{3x^2}$

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Illustration 20

Find differentiation of x^x w. r. t. $\cos x$

Solution:

$$\Rightarrow \frac{d(x^x)}{d(\cos x)} = \frac{x^x(\ell nx+1)}{-\sin x}$$

Illustration 21

Differentiate $\sin^{-1} x$ w. r. t. e^x

Solution:

$$\frac{d\left(\sin^{-1} x\right)}{d\left(e^x\right)} = \left(\frac{\frac{1}{\sqrt{1-x^2}}}{e^x}\right)$$

Illustration 22

Differentiation of $e^{\sin^{-1}(x)}$ w.r.t $e^{-\cos^{-1}(x)}$.

Solution:

$$\frac{d\left(e^{\sin^{-1} x}\right)}{d\left(e^{-\cos^{-1} x}\right)} \Rightarrow \frac{e^{\sin^{-1} x}.\frac{1}{\sqrt{1-x^2}}}{e^{-\cos^{-1} x}.\left(\frac{-1(-1)}{\sqrt{1-x^2}}\right)} \Rightarrow e^{\frac{\pi}{2}} \qquad \left\{\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}\right\}$$

Alternate:

$$\frac{d\left(e^{\sin^{-1} x}\right)}{d\left(e^{-\cos^{-1} x}\right)} = \frac{d\left(e^{\frac{\pi}{2}-\cos^{-1} x}\right)}{d\left(e^{-\cos^{-1} x}\right)} \Rightarrow \frac{d\left(e^{\frac{\pi}{2}}.e^{-\cos^{-1} x}\right)}{d\left(e^{-\cos^{-1} x}\right)} = e^{\frac{\pi}{2}} \; .1 = e^{\pi/2}$$

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Illustration 23

Differentiate $f(\tan x)$ w.r.t. to $g(\sec x)$ at $x = \frac{\pi}{4}$. Where $f'(1) = 3$ and $g'\left(\sqrt{2}\right) = 4$

Solution:

$$\frac{d\left(f\left(\tan x\right)\right)}{d\left(g\left(\sec x\right)\right)} = \frac{\left(f'\left(\tan x\right)\right).\sec^2 x}{\left(g'\left(\sec x\right)\right).\sec x \tan x} = \frac{f'\left(\tan\frac{\pi}{4}\right).\sec^2\frac{\pi}{4}}{g'\left(\sec\frac{\pi}{4}\right).\tan\frac{\pi}{4}.\sec\frac{\pi}{4}} = \frac{3.2}{4\times\sqrt{2}} = \frac{3}{2\sqrt{2}}$$

6. PARAMETRIC DIFFERENTIATION:

If $x = f(t)$ & $y = g(t)$, then $\frac{dy}{dx} = \frac{dy}{dt}.\frac{1}{dx/dt} = \frac{g\prime(t)}{f\prime(t)}$

Where $\frac{dx}{dt} = f'(t)$ and $\frac{dy}{dt} = g'(t)$

Illustration 24

If $x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2at}{1+t^2}$, then $\frac{dy}{dx} =$

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Solution:

$x = \frac{1-t^2}{1+t^2}$ and $y = \frac{2at}{1+t^2}$

Differentiating with respect to t, we get

$$\frac{dx}{dt} = \frac{(1+t^2)(0-2t)-(1-t^2)(0+2t)}{(1+t^2)^2} = -\frac{4t}{(1+t^2)^2}$$

and $\frac{dy}{dt} = \frac{(1+t^2)2a-2at(2t)}{(1+t^2)^2} = \frac{2a(1-t^2)}{(1+t^2)^2}$

$$\Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{a(1-t^2)}{-2t}$$

$$\Rightarrow \frac{dy}{dx} = \frac{a(t^2-1)}{2t}.$$

Illustration 25

If $\tan y = \frac{2t}{1-t^2}$ and $\sin x = \frac{2t}{1+t^2}$, then find

Solution:

$\tan y = \frac{2t}{1-t^2}$ ...(i)

and $\sin x = \frac{2t}{1+t^2}$ ...(ii)

From (i), differentiating *w.r.t.* t of y, we get, $\sec^2 y \frac{dy}{dx} = \frac{2(1+t^2)}{(1-t^2)^2}$ and $\frac{dy}{dt} = \frac{2(1+t^2)}{(1-t^2)^2} \cdot \frac{1}{(1+\tan^2 y)}$

or $\frac{dy}{dt} = \frac{2(1+t^2)}{(1-t^2)^2} \cdot \frac{1}{\left[1+\left(\frac{2t}{1-t^2}\right)^2\right]} = \frac{2}{1+t^2}$ ...(iii)

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and from (ii), differentiating w.r.t. t of x, we get

$\cos x \frac{dx}{dt} = \frac{2(1-t^2)}{(1+t^2)^2}$

or $\frac{dx}{dt} = \frac{2(1-t^2)}{(1+t^2)^2} \frac{1}{\sqrt{1-\frac{(2t)^2}{(1+t^2)^2}}} = \frac{2}{1+t^2}$ ...(iv)

Hence $\frac{dy}{dx} = 1$.

Illustration 26

If $x = at^2, y = 2at$, then find $\frac{d^2y}{dx^2}$

Solution:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{2a}{2at} \Rightarrow \frac{dy}{dx} = \frac{1}{t} = \frac{2a}{y}$$

$$\Rightarrow y\frac{dy}{dx} = 2a \Rightarrow y\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^2 = 0$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{-(dy/dx)^2}{y} = -\frac{1}{2at^3}.$$

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Illustration 27

If $x = 2\cos t - \cos 2t$, $y = 2\sin t - \sin 2t$, then at $t = \frac{\pi}{4}$, $\frac{dy}{dx}$ =?

Solution:

$\frac{dx}{dt} = -2\sin t + 2\sin 2t$ and $\frac{dy}{dt} = 2\cos t - 2\cos 2t$

$\Rightarrow \frac{dy}{dx} = \frac{\cos t - \cos 2t}{\sin 2t - \sin t}$

Put $t = \frac{\pi}{4}$, we have $\left[\frac{dy}{dx}\right]_{t=\pi/4}$

$= \frac{\cos \pi/4 - \cos \pi/2}{\sin \pi/2 - \sin \pi/4} = \sqrt{2} + 1.$

7. DERIVATIVE OF IMPLICIT FUNCTIONS:

$\phi(x, y) = 0$

(a) To find $\frac{dy}{dx}$ of implicit functions, we differentiate each term w.r.t. x regarding y as a function of x then collect terms with $\frac{dy}{dx}$ together on one side.

(b) In the case of implicit functions, generally, both x & y are present in answers of $\frac{dy}{dx}$. Corresponding to every curve represented by an implicit equation, there exist one or more explicit functions representing that equation. It can be shown that $\frac{dy}{dx}$ at any point on the curve remains the same whether the process of differentiation is done explicitly or implicitly.

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Illustration 28

If $\sin y = x\sin(a + y)$ then prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$. Also find $\frac{dy}{dx}$ explicitly.

Solution:

$x = \frac{\sin y}{\sin(a+y)} \Rightarrow \frac{dx}{dy} = \frac{\sin(a+y)\cos y - \sin y\cos(a+y)}{\sin^2(a+y)}$

$\Rightarrow \frac{dx}{dy} = \frac{\sin a}{\sin^2(a+y)}$

Explicitly $\frac{1}{x} = \frac{\sin(a+y)}{\sin y} = \sin a \cdot \cot y + \cos a \Rightarrow \sin a \cdot \cot y = \frac{1}{x} - \cos a$

$\Rightarrow \cot y = \frac{1 - x\cos a}{x\sin a}$

$\Rightarrow -\text{cosec}^2 y \frac{dy}{dx} = -\frac{1}{x^2 \sin a}$

$\Rightarrow \frac{dy}{dx} = \frac{1}{x^2 \sin a(1 + \cot^2 y)}$

(substitute the value of cot y to get $\frac{dy}{dx}$ explicitly)]

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Illustration 29

If $x\sqrt{1+y}+y\sqrt{1+x}=0$, then prove that $\frac{dy}{dx}=-\frac{1}{(1+x)^2}$ (explicit form)

Solution:

$\left(x\sqrt{1+y}\right)^2=\left(y\sqrt{1+x}\right)^2 \Rightarrow xy(x-y)+(x^2-y^2)=0$ or $(x-y)(xy+x+y)=0$

$\therefore y=-\frac{x}{1+x}=\frac{1}{1+x}-1$ (as $y=x$ does not satisfy)

hence $y'=-\frac{1}{(1+x)^2}$

Illustration 30

If $\cos(x+y)=y\sin x$, then find $\frac{dy}{dx}$

Solution:

$\cos(x+y)=(y\sin x)$

$\Rightarrow -\sin(x+y)\left(1+\frac{dy}{dx}\right)=y\cos x+\sin x\frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx}=-\frac{y\cos x+\sin(x+y)}{\sin(x+y)+\sin x}$.

Illustration 31

If $\sin^2 x+2\cos y+xy=0$, then find $\frac{dy}{dx}$

Solution:

$\sin^2 x+2\cos y+xy=0$

$\Rightarrow 2\sin x\cos x-2\sin y\frac{dy}{dx}+y+x\frac{dy}{dx}=0$

$\therefore \frac{dy}{dx}=\frac{y+\sin 2x}{2\sin y-x}$.

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Illustration 32

If $x^3+8xy+y^3=64$, then find $\frac{dy}{dx}$

Solution:

$x^3+8xy+y^3=64 \Rightarrow 3x^2+8\left(y+x\frac{dy}{dx}\right)+3y^2\frac{dy}{dx}=0$

$\therefore \frac{dy}{dx}=-\frac{3x^2+8y}{8x+3y^2}$.

8. DERIVATIVE OF INFINITE SERIES:

Illustration 33

If $y=e^{x+e^{x+e^{x+\ldots\infty}}}$, then $\frac{dy}{dx}=?$

Solution:

$y=e^{x+y}$

$\Rightarrow \log y=(x+y)\log e \Rightarrow \frac{1}{y}\frac{dy}{dx}=1+\frac{dy}{dx}$

$\Rightarrow \frac{dy}{dx}=\frac{y}{1-y}$.

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Illustration 34

If $y=\sqrt{\log x+\sqrt{\log x+\sqrt{\log x+\ldots\ldots\infty}}}$, then $\frac{dy}{dx}=?$

Solution:

$y=\sqrt{\log x+y}$

$\Rightarrow y^2=\log x+y$

$\Rightarrow 2y\frac{dy}{dx}=\frac{1}{x}+\frac{dy}{dx} \Rightarrow \frac{dy}{dx}=\frac{1}{x(2y-1)}$

Illustration 35

If $y=(\sin x)^{(\sin x)^{(\sin x)\ldots\ldots\infty}}$, then $\frac{dy}{dx}=?$

Solution:

$y=(\sin x)^{(\sin x)^{(\sin x)\ldots\ldots\infty}}$

$\Rightarrow y=(\sin x)^y \Rightarrow \log_e y=y\log\sin x \Rightarrow \frac{1}{y}\frac{dy}{dx}=\frac{dy}{dx}[\log\sin x+y\cot x]$

$\therefore \frac{dy}{dx}=\frac{y^2\cot x}{1-y\log\sin x}$.

Illustration 36

The differential equation satisfied by the function $y=\sqrt{\sin x+\sqrt{\sin x+\sqrt{\sin x+\ldots\ldots\infty}}}$, is

Solution:

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$y=\sqrt{\sin x+\sqrt{\sin x+\sqrt{\sin x+\ldots\ldots\infty}}}$

$\Rightarrow y=\sqrt{\sin x+y}$

$\Rightarrow y^2=\sin x+y$

On differentiating both sides, we get

$2y\frac{dy}{dx}=\cos x+\frac{dy}{dx} \Rightarrow \frac{dy}{dx}(2y-1)=\cos x$

9. DERIVATIVE OF HOMOGENEOUS EQUATION:

Illustration 37

If $ax^2+2hxy+by^2=0$, then prove that $\frac{dy}{dx}=-\frac{ax+hy}{hx+by}=\frac{y}{x}$

Solution:

$2ax+2by\,y'+2h(xy'+y)=0 \Rightarrow (ax+hy)+(by+hx)y'=0$

$\Rightarrow y'=-\frac{ax+hy}{by+hx}$

$\because\ ax^2+hxy+hxy+by^2=0 \Rightarrow x(ax+hy)+y(by+hx)=0$

$\Rightarrow -\frac{ax+hy}{by+hx}=\frac{y}{x}$

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Illustration 38

If $\sqrt{1-x^2}+\sqrt{1-y^2}=a(x-y)$ then prove that

(i) $\frac{dy}{dx}$ is independent of a

(ii) $\frac{dy}{dx}=\sqrt{\frac{1-y^2}{1-x^2}}$

Solution:

Put $x=\sin\theta\ x, y\in[-1,1]$

And $y=\sin\phi \qquad \theta\ \&\ \phi\in\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$

$\cos\theta+\cos\phi=a(\sin\theta-\sin\phi)$

$\therefore\ \tan\left(\frac{\theta-\phi}{2}\right)=\frac{1}{a}\Rightarrow\frac{\theta-\phi}{2}=\tan^{-1}\left(\frac{1}{a}\right)$

$\sin^{-1}x-\sin^{-1}y=2\tan^{-1}\left(\frac{1}{a}\right);\ \therefore\ \frac{dy}{dx}=\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$

Illustration 39

If $x^m.y^n=(x+y)^{m+n}$ then prove that $\frac{dy}{dx}$ is independent of m and n.

Solution:

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$m\ell nx+n\ell ny=(m+n)\ell n(x+y)\Rightarrow\frac{m}{x}+\frac{n}{y}.y'=\frac{m+n}{x+y}(1+y')$

$\left(\frac{m}{x}-\frac{m+n}{x+y}\right)=y'\left(\frac{m+n}{x+y}-\frac{n}{y}\right)$ or $\frac{my-nx}{x}=\frac{dy}{dx}\left[\frac{my-nx}{y}\right]$

Hence $\frac{dy}{dx}=\frac{y}{x}$

10. DERIVATIVE OF INVERSE FUNCTION:

Theorem:

If the inverse functions f & g are defined by $y=f(x)$ & $x=g(y)$ & if $f'(x)$ exists & $f'(x)\neq 0$, then $g'(y)=\frac{1}{f'(x)}$. This result can also be written as, if $\frac{dy}{dx}$ exists & $\frac{dy}{dx}\neq 0$, then $\frac{dx}{dy}=\frac{1}{\frac{dy}{dx}}$ or $\frac{dy}{dx}.\frac{dx}{dy}=1$

Proof: $g(x)=f^{-1}(x)$

$\Rightarrow f(g(x))=x$ differentiate w.r.t. x

$\Rightarrow f'(g(x)).g'(x)=1\Rightarrow\ \boxed{f'(g(x))=\frac{1}{g'(x)}}$

Similarly, $\boxed{g'(f(x))=\frac{1}{f'(x)}}$

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Illustration 40

Let $g(x) = f^{-1}(x)$ where $f(x) = x^3 + 3x$, then find $f^{-1}(4)$ or $g(4)$.

Solution:

$g(4) = f^{-1}(4) = k$

$f^{-1}(4) = k$

$f(k) = 4$

$k^3 + 3k = 4$

$k^3 + 3k - 4 = 0$

$k = 1 g(4) = 1$

Illustration 41

Let $g(x) = f^{-1}(x)$ where $f(x) = x + e^x$ then find $f^{-1}(1)$.

Solution:

$g(1) = f^{-1}(1) = k$

$f(k) = 1$

$k + e^k = 1$

$k = 0$

$\therefore g(1) = 0$

$\therefore f^{-1}(1) = 0$

Illustration 42

Let $f(x) = \exp(x^3 + x^2 + x)$ for any real number x and let g be the inverse function for f. The value of $g'(e^3)$ is –

(A) $\frac{1}{6e^3}$ (B) $\frac{1}{6}$ (C) $\frac{1}{34e^{39}}$ (D) 6

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Ans. (A)

Solution:

$y = e^{x^3+x^2+x}$

$\Rightarrow \frac{dy}{dx} = e^{x^3+x^2+x}.(3x^2 + 2x + 1) \Rightarrow g'(y) = \frac{dx}{dy} = \frac{1}{e^{x^3+x^2+x}.(3x^2+2x+1)}$

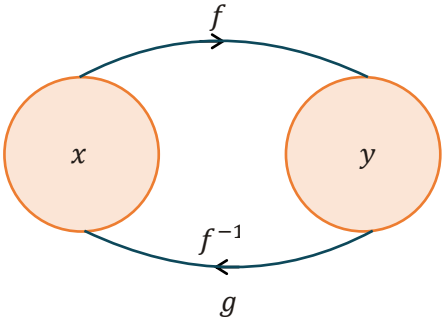
$\Rightarrow y = e^3 \Rightarrow e^3 = e^{x^3+x^2+x} \Rightarrow x^3 + x^2 + x - 3 = 0$

$\Rightarrow (x-1)(x^2 + 2x + 3) = 0 \Rightarrow x = 1$

$\therefore g'(e^3) = \frac{1}{6e^3}$

Illustration 43

If $y = f(x) = x^3 + x^5$ and g is the inverse of f find $g'(2)$



$y = f(x);\ x = g(y)$

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Solution:

$\frac{dy}{dx} = 3x^2 + 5x^4 \Rightarrow g'(y) = \frac{dx}{dy} = \frac{1}{3x^2+5x^4}$

when $y = 2$ then $2 = x^3 + x^5 \Rightarrow x = 1$

$\therefore g'(2) = \frac{dx}{dy}\Big]_{\substack{x=1 \\ y=2}} = \frac{1}{3+5} = \frac{1}{8}$

Alternatively: $(gof)(x) = x$

$g'[f(x)]\, f'(x) = 1$; when $f(x) = 2$ then $x = 1$

$g'(2) \cdot f'(1) = 1$ (but $f'(1) = 8$)

$\therefore g'(2) = \frac{1}{8}$

11. DERIVATIVES OF INVERSE TRIGONOMETRIC FUNCTIONS BY TRANSFORMING THEM INTO SIMPLER FUNCTIONS:

Standard Substitution:

(a) For terms of the form $x^2 + a^2$ or $\sqrt{x^2 + a^2}$, put $x = a\tan\theta$ or $a\cot\theta$.

(b) For terms of the form $x^2 - a^2$ or $\sqrt{x^2 - a^2}$, put $x = a\sec\theta$ or $a\,\text{cosec}\theta$.

For terms of the form $a^2 - x^2$ or $\sqrt{a^2 - x^2}$, put $x = a\sin\theta$ or $a\cos\theta$.

(c) If both $\sqrt{a+x}, \sqrt{a-x}$ are present, then put $x = a\cos 2\theta$.

For the type $\sqrt{(x-a)(b-x)}$ or $\sqrt{\frac{x-a}{b-x}}$ put $x = a\cos^2\theta + b\sin^2\theta$.

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(d) For the type $\sqrt{(x-a)(x-b)}$ or $\sqrt{\frac{x-a}{x-b}}$ put $x = a\sec^2\theta - b\tan^2\theta$.

(e) For the type $(\sqrt{x^2 + a^2} \pm x)^n$ or $(x \pm \sqrt{x^2 - a^2})^n$, put the expression within the bracket = t.

Note that $\int \sqrt{a^2 + x^2}\,dx$ & $\int \sqrt{x^2 - a^2}\,dx$ to be evaluated by parts

Illustration 44

Let $y = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), x \in (0,1)$

Solution:

Let $\tan^{-1} x = \theta, \theta \in \left(0, \frac{\pi}{4}\right)$

$x = \tan\theta$

$y = \cos^{-1}\left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right) = \cos^{-1}\cos 2\theta, 2\theta \in \left(0, \frac{\pi}{2}\right) = 2\theta$

$y = 2\tan^{-1} x$

$\frac{dy}{dx} = \frac{2}{1+x^2}$

Illustration 45

If $f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find

(i) $f'(2)$ (ii) $f'\left(\frac{1}{2}\right)$ (iii) $f'(1)$

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Solution:

$x = \tan\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2} \Rightarrow y = \sin^{-1}\sin 2\theta$

$$y = \begin{cases} \pi - 2\theta & \frac{\pi}{2} < 2\theta < \pi \\ 2\theta & \frac{-\pi}{2} \le 2\theta \le \frac{\pi}{2} \\ -(\pi + 2\theta) & -\pi < 2\theta < -\frac{\pi}{2} \end{cases} \Rightarrow f(x) = \begin{cases} \pi - 2\tan^{-1}x & x > 1 \\ 2\tan^{-1}x & -1 \le x \le 1 \\ -(\pi + 2\tan^{-1}x) & x < -1 \end{cases} \Rightarrow f'(x) = \begin{cases} -\frac{2}{1+x^2} & x > 1 \\ \frac{2}{1+x^2} & -1 < x < 1 \\ \frac{-2}{1+x^2} & x < -1 \end{cases}$$

(i) $f'(2) = -\frac{2}{5}$ (ii) $f'\left(\frac{1}{2}\right) = \frac{8}{5}$ (iii) $f'(1^+) = -1$ and $f'(1^{-1}) = +1 \Rightarrow f'(1)$ does not exist

Illustration 46

$$\frac{d}{dx}\left\{\sin^2\left(\cot^{-1}\sqrt{\frac{1+x}{1-x}}\right)\right\}$$

Solution:

Let $y = \sin^2\left(\cot^{-1}\sqrt{\frac{1+x}{1-x}}\right)$. Put $x = \cos 2\theta$ $\theta \in \left(0, \frac{\pi}{2}\right]$

$\therefore y = \sin^2\cot^{-1}\left(\sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}}\right) = \sin^2\cot^{-1}(\cot\theta)$

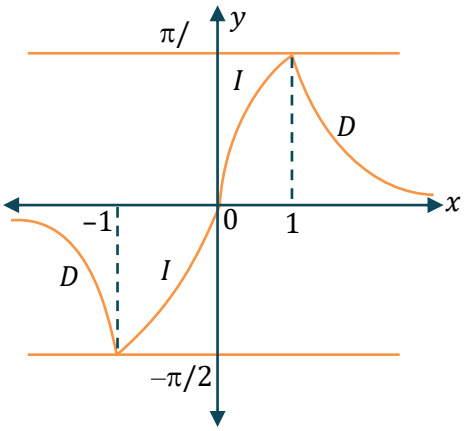
$\therefore y = \sin^2\theta = \frac{1-\cos 2\theta}{2} = \frac{1-x}{2} = \frac{1}{2} - \frac{x}{2}$

$\therefore \frac{dy}{dx} = -\frac{1}{2}$.

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12. ANALYSIS AND GRAPHS OF SOME INVERSE TRIGONOMETRIC FUNCTIONS:

(a) $y = f(x) = \sin^{-1}\left(\frac{2x}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & |x| \le 1 \\ \pi - 2\tan^{-1}x & x > 1 \\ -(\pi + 2\tan^{-1}x) & x < -1 \end{cases}$



Important points :

(i) Domain is $x \in R$ & range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(ii) f is continuous for all x but not differentiable at $x = 1, -1$

(iii) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for } |x| < 1 \\ \text{non existent} & \text{for } |x| = 1 \\ \frac{-2}{1+x^2} & \text{for } |x| > 1 \end{cases}$

(iv) Increasing in $[-1, 1]$ & Decreasing in $(-\infty, -1], [1, \infty)$

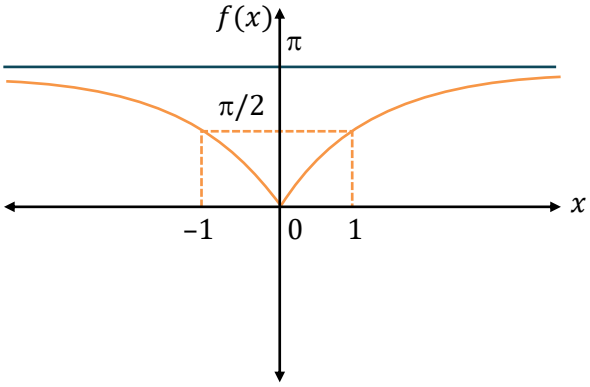
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(b) Consider $y = f(x) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = \begin{cases} 2\tan^{-1}x & \text{if} \quad x \geq 0 \\ -2\tan^{-1}x & \text{if} \quad x < 0 \end{cases}$

Important points :

(i) Domain is $x \in R$ & range is $[0, \pi)$

(ii) Continuous for all x but not differentiable at $x = 0$



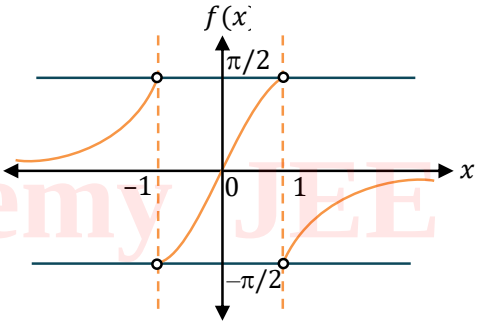
(iii) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & \text{for} \quad x > 0 \\ \text{non existent} & \text{for} \quad x = 0 \\ -\frac{2}{1+x^2} & \text{for} \quad x < 0 \end{cases}$

(iv) Increasing in $[0, \infty)$ & Decreasing in $(-\infty, 0]$

(c) $y = f(x) = \tan^{-1}\frac{2x}{1-x^2} = \begin{cases} 2\tan^{-1}x & |x| < 1 \\ \pi + 2\tan^{-1}x & x < -1 \\ -(\pi - 2\tan^{-1}x) & x > 1 \end{cases}$

Important points :

(i) Domain is $R - \{1, -1\}$ & range is $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



(ii) $\frac{dy}{dx} = \begin{cases} \frac{2}{1+x^2} & |x| \neq 1 \\ \text{non existent} & |x| = 1 \end{cases}$

(iii) It is bounded for all x

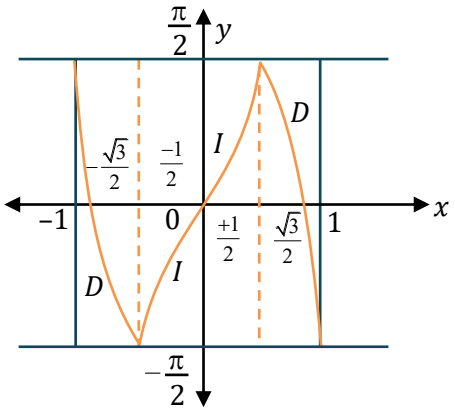
(d) $y = f(x) = \sin^{-1}(3x - 4x^3) = \begin{cases} -(\pi + 3\sin^{-1}x) & \text{if} \quad -1 \leq x < -\frac{1}{2} \\ 3\sin^{-1}x & \text{if} \quad -\frac{1}{2} \leq x \leq \frac{1}{2} \\ \pi - 3\sin^{-1}x & \text{if} \quad \frac{1}{2} < x \leq 1 \end{cases}$

Important points :

(i) Domain is $x \in [-1, 1]$ & range is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

(ii) Continuous everywhere in its domain

(iii) Not derivable at $x = -\frac{1}{2}, \frac{1}{2}, 1, -1$



(iv) $\frac{dy}{dx} = \begin{cases} \frac{3}{\sqrt{1-x^2}} & \text{if} \quad x \in (-\frac{1}{2}, \frac{1}{2}) \\ -\frac{3}{\sqrt{1-x^2}} & \text{if} \quad x \in (-1, -\frac{1}{2}) \cup (\frac{1}{2}, 1) \end{cases}$

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(v) Increasing in $\left[-\frac{1}{2},\frac{1}{2}\right]$ and Decreasing in $\left[-1,-\frac{1}{2}\right],\left[\frac{1}{2},1\right]$

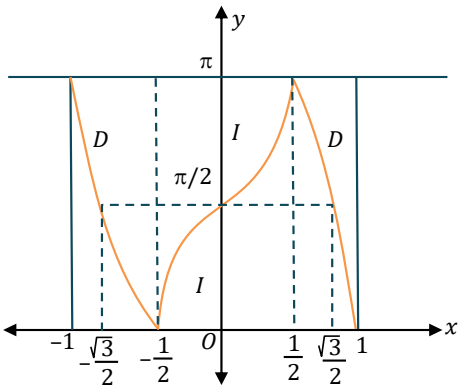
(e) $y=f(x)=\cos^{-1}(4x^3-3x)=\begin{cases}3\cos^{-1}x-2\pi & \text{if } -1\le x<-\frac{1}{2}\\ 2\pi-3\cos^{-1}x & \text{if } -\frac{1}{2}\le x\le\frac{1}{2}\\ 3\cos^{-1}x & \text{if } \frac{1}{2}\le x\le 1\end{cases}$

Important points :

(i) Domain is $x\in[-1,\ 1]$ & range is $[0,\ \pi]$

(ii) Continuous everywhere in its domain

(iii) Not derivable at $x=-\frac{1}{2},\ \frac{1}{2},\ 1,\ -1$



(iv) $\frac{dy}{dx}=\begin{cases}\frac{3}{\sqrt{1-x^2}} & \text{if } x\in\left(-\frac{1}{2},\frac{1}{2}\right)\\ -\frac{3}{\sqrt{1-x^2}} & \text{if } x\in\left(-1,-\frac{1}{2}\right)\cup\left(\frac{1}{2},1\right)\end{cases}$

(v) Increasing in $\left[-\frac{1}{2},\frac{1}{2}\right]$ &

Decreasing in $\left[-1,-\frac{1}{2}\right],\left[\frac{1}{2},1\right]$

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Illustration 47

Obtain differential coefficient of $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$ with respect to $\cos^{-1}\sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

Solution:

Assume $u=\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}$, $v=\cos^{-1}\sqrt{\frac{1+\sqrt{1+x^2}}{2\sqrt{1+x^2}}}$

The function needs simplification before differentiation Let $x=\tan\theta$; $\theta\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

$$\therefore\ u=\tan^{-1}\left(\frac{\sec\theta-1}{\tan\theta}\right)=\tan^{-1}\left(\frac{1-\cos\theta}{\sin\theta}\right)=\tan^{-1}\left(\tan\frac{\theta}{2}\right)=\frac{\theta}{2}$$

$$v=\cos^{-1}\sqrt{\frac{1+\sec\theta}{2\sec\theta}}=\cos^{-1}\sqrt{\frac{1+\cos\theta}{2}}=\cos^{-1}\left(\cos\frac{\theta}{2}\right)=\begin{cases}\frac{\theta}{2}, & \theta\in\left(0,\frac{\pi}{2}\right)\\ -\frac{\theta}{2}, & \theta\in\left(-\frac{\pi}{2},0\right)\end{cases}$$

$\Rightarrow u=v$ or $-v$

$$\Rightarrow\frac{du}{dv}=\begin{cases}1, & x>0\\ -1, & x<0\end{cases}$$

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13. SUCCESSIVE DIFFERENTIATION:

$y = f(x)$ the popular symbols used to denote the derivatives are $\frac{dy}{dx} = Dy = f'(x) = y_1 = y'$.

Higher order derivative are denoted as $\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} = D^2y = f''(x) = y_2$ or y'' etc.

Similarly, n^{th} order derivatives are denoted by $\frac{d^ny}{dx^n}$ or D^ny or y_n

Illustration 48

$f(x) = be^{ax} + ae^{bx}$ find $f''(0)$

Solution:

$f'(x) = be^{ax}.a + ae^{bx}.b \Rightarrow f''(x) = a^2be^{ax} + ab^2e^{bx}$

$f''(0) = a^2b.1 + ab^2.1$

$= a^2b + ab^2$

$= ab\,(a + b)$

Illustration 49

$y = \log|x|$ find (i) $\frac{dy}{dx}$, (ii) $\frac{d^2y}{dx^2}$

Solution:

(i) $y = \log|x|$

$x > 0$: $y = \log x$, $\frac{dy}{dx} = \frac{1}{x}$

$x < 0$: $y = \log(-x)$, $\frac{dy}{dx} = \frac{1}{(-x)} \times (-1) = \frac{1}{x}$

(ii) $\frac{d^2y}{dx^2} = \frac{-1}{x^2}$

Illustration 50

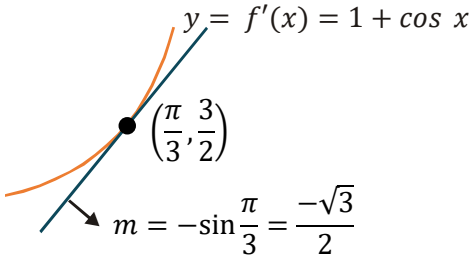
If $y = x + \sin x$, then find

(i) $\frac{d^2y}{dx^2}$ (ii) equation of tangent of curve $y = f'(x)$ at $x = \frac{\pi}{3}$

Solution:

(i) $\frac{dy}{dx} = 1 + \cos x \Rightarrow \frac{d^2y}{dx^2} = -\sin x$

(ii) at $x = \frac{\pi}{3}$



equation: $y - \frac{3}{2} = -\frac{\sqrt{3}}{2}\left(x - \frac{\pi}{3}\right)$

Illustration 51

If $y = \left(\frac{1}{x}\right)^x$, then prove $\frac{d^2y}{dx^2}$ at $x = 1$ is 0.

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Solution:

$\Rightarrow \log y = x\log\frac{1}{x} \Rightarrow \log y = -x\log x \Rightarrow \frac{dy}{dx} = -y(1+\ell n\, x)$

$\Rightarrow \frac{d^2y}{dx^2} = \frac{-dy}{dx}(1+\ell n\, x) - y\left(0+\frac{1}{x}\right) = y(1+\ell nx)^2 - y.\frac{1}{x}$

{at $x = 1 \Rightarrow y = 1$}

$\Rightarrow \left(\frac{d^2y}{dx^2}\right)_{x=1} = 1(1+\ell n1)^2 - 1.\frac{1}{1} = 1(1) - 1 = 1-1 = 0$

Illustration 52

$y = \sin^{-1} x$, show that $(1-x^2)\frac{d^2y}{dx^2} - \frac{xdy}{dx} = 0$

Solution:

$y = \sin^{-1} x$

$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} \Rightarrow \sqrt{1-x^2}\frac{dy}{dx} = 1$

$\frac{d}{dx}\left(\sqrt{1-x^2}\frac{dy}{dx}\right) = 0 \Rightarrow \sqrt{1-x^2}.\frac{d^2y}{dx^2} + \frac{dy}{dx}\frac{d}{dx}\left(\sqrt{1-x^2}\right)$

$\sqrt{1-x^2}.\frac{d^2y}{dx^2} - \frac{dy}{dx}\frac{2x}{2\sqrt{1-x^2}} = 0$

$(1-x^2)\frac{d^2y}{dx^2} - \frac{xdy}{dx} = 0$

Hence proved.

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14. DEDUCTION OF NEW IDENTITIES BY DIFFERENTIATING A GIVEN IDENTITY:

Illustration 53

If $\cos\frac{x}{2}.\cos\frac{x}{2^2}.\cos\frac{x}{2^3}.....\cos\frac{x}{2^n} = \frac{\sin x}{2^n\sin(x/2^n)}$, then prove that $\sum_{r=1}^{n}\frac{1}{2^r}\tan\frac{x}{2^r} = \frac{1}{2^n}\cot\frac{x}{2^n} - \cot x$

Solution:

$\cos\frac{x}{2}.\cos\frac{x}{2^2}.\cos\frac{x}{2^3}.....\cos\frac{x}{2^n} = \frac{\sin x}{2^n\sin(x/2^n)}$

Take log:

$\log\cos\frac{x}{2} + \log\cos\frac{x}{2^2} + \log\cos\frac{x}{2^3}.....+\log\cos\frac{x}{2^n} = \log\sin x - \log 2^n - \log\sin\frac{x}{2^n}$

Differentiate w.r.t x

$\Rightarrow \frac{1}{2}\tan\frac{x}{2} + \frac{1}{2^2}\tan\frac{x}{2^2} + ...+\frac{1}{2^n}\tan\frac{x}{2^n} = \frac{1}{2^n}\cot\frac{x}{2^n} - \cot x$

$\Rightarrow \boxed{\sum_{r=1}^{n}\frac{1}{2^r}\tan\frac{x}{2^r} = \frac{1}{2^n}\cot\frac{x}{2^n} - \cot x}$

Illustration 54

$(1+x)^n = C_0 + C_1x + C_2x^2 + ... + C_nx^n$, prove that

(i) $C_1 + 2C_2 + 3C_3 + ... + nC_n = n2^{n-1}$

(ii) $C_0 + 2C_1 + 3C_2 + ... + (n+1)C_n = (n+2)2^{n-1}$

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Solution:

(i) differentiate

$n(1+x)^{n-1} = C_1 + 2C_2x + 3C_3x^2 + \ldots\ldots\ldots + nC_n.x^{n-1}$

Put $x = 1$

$n.2^{n-1} = C_1 + 2C_2 + 3C_3 + \ldots\ldots\ldots + nC_n$

(ii) Multiply both side by x

$x(1+x)^n = C_0x + C_1x^2 + C_n x^{n+1}$

Now, differentiate.

$(1+x)^n + x.n(1+x)^{n-1} = c_0 + 2c_1x + \ldots (n+1)c_nx^n$

Put $x = 1$

$2^n + n.2^{n-1} = c_0 + 2c_1 + 3c_2 + \ldots\ldots\ldots + (n+1)c_n$

$\Rightarrow 2^{n-1}(n+2) = c_0 + 2c_1 + 3c_2 + \cdots + (n+1)c_n$

Hence proved.

15. DERIVATIVE OF FUNCTIONS EXPRESSED IN THE DETERMINANT FORM:

Let $F(x) = \begin{vmatrix} f & g & h \\ u & v & w \\ \ell & m & n \end{vmatrix}$ where all functions are differentiable then

$$F'(x) = \begin{vmatrix} f' & g' & h' \\ u & v & w \\ \ell & m & n \end{vmatrix} + \begin{vmatrix} f & g & h \\ u' & v' & w' \\ \ell & m & n \end{vmatrix} + \begin{vmatrix} f & g & h \\ u & v & w \\ \ell' & m' & n' \end{vmatrix}$$

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The same operation can also be done column wise.

Proof: Differentiation of Determinant –

$\Rightarrow P(x) = \begin{vmatrix} f(x) & g(x) \\ h(x) & r(x) \end{vmatrix} \Rightarrow P(x) = f(x)r(x) - g(x)h(x)$

$\Rightarrow P'(x) = f'(x)\,r(x) + f(x)\,r'(x) - g'(x)h(x) - g(x)\,h'(x)$

$\Rightarrow P'(x) = \begin{vmatrix} f'(x) & g'(x) \\ h(x) & r(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) \\ h'(x) & r'(x) \end{vmatrix}$

Illustration 55

$f(x) = \begin{vmatrix} e^x & x^2+x+1 & \sin x \\ x^2 & \cos x & 1 \\ x^3+x & \tan x & 3x^2+1 \end{vmatrix}$, find $f'(0)$

Solution:

$$f'(x) = \begin{vmatrix} e^x & 2x+1 & \cos x \\ x^2 & \cos x & 1 \\ x^3+x & \tan x & 3x^2+1 \end{vmatrix} + \begin{vmatrix} e^x & x^2+x+1 & \sin x \\ 2x & -\sin x & 0 \\ x^3+x & \tan x & 3x^2+1 \end{vmatrix} + \begin{vmatrix} e^x & x^2+x+1 & \sin x \\ x^2 & \cos x & 1 \\ 3x^3+1 & \sec^2 x & 6x \end{vmatrix}$$

$= 1 + 0 - 1 + 1 = 1$

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16. L' HOSPITAL'S RULE

(a) This rule is applicable for the indeterminate forms of the type $\frac{0}{0}, \frac{\infty}{\infty}$. If the function $f(x)$ and $g(x)$ are differentiable in certain neighborhood of the point $'a'$, except, may be, at the point $'a'$ itself and $g'(x) \neq 0$, and if $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = 0$ **or** $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = \infty$, then $\lim_{x\to a} \frac{f(x)}{g(x)} = \lim_{x\to a} \frac{f'(x)}{g'(x)}$ provided the limit $\lim_{x\to a} \frac{f'(x)}{g'(x)}$ exists (L' Hospital's rule). The point $'a'$ may be either finite or improper $(+\infty$ or $-\infty)$.

(b) Indeterminate forms of the type $0.\infty$ or $\infty - \infty$ are reduced to forms of the type $\frac{0}{0}$ or $\frac{\infty}{\infty}$ by algebraic transformations.

(c) Indeterminate forms of the type $1^{\infty}, \infty^0$ or 0^0 are reduced to forms of the type $0 \times \infty$ by taking logarithms or by the transformation $[f(x)]^{\phi(x)} = e^{\phi(x).\ell n f(x)}$

Illustration 56

Evaluate $\lim_{x\to 0} |x|^{\sin x}$

Solution:

$\lim_{x\to 0} |x|^{\sin x} = \lim_{x\to 0} e^{\sin x \log_e |x|} = e^{\lim \frac{\log_e |x|}{\cos ecx}}$

$= e^{\lim_{x\to 0-} \frac{1/x}{\text{cosec}\, x \cot x}}$ (applying L'Hopital's rule)

$= e^{\lim_{x\to 0-} \frac{\sin^2 x}{x\cos x}} = e^{\lim_{x\to 0-} \left(\frac{\sin x}{x}\right)^2 \cdot \left(\frac{x}{\cos x}\right)} = e^{-(1)^2.(0)} = e^0 = 1$

Illustration 57

Solve $\lim_{x\to 0^+} \log_{\sin x} \sin 2x$.

Solution:

Here $\lim_{x\to 0^+} \log_{\sin x} \sin 2x$

$= \lim_{x\to 0^+} \frac{\log \sin 2x}{\log \sin x}$ $\left(\frac{-\infty}{-\infty} \text{form}\right)$

$= \lim_{x\to 0^+} \frac{\frac{1}{\sin 2x} \cdot 2\cos 2x}{\frac{1}{\sin x} \cdot \cos x}$ {applying L'Hopital's rule}

$= \lim_{x\to 0^+} \frac{\left(\frac{(2x)}{\sin(2x)}\right)\cos 2x}{\left(\frac{x}{\sin x}\right)\cos x} = \lim_{x\to 0^+} \frac{\cos 2x}{\cos x} = 1$

Illustration 58

Evaluate $\lim_{n\to\infty} \left(\frac{e^n}{\pi}\right)^{1/n}$.

Solution:

Here, $A = \lim_{n\to\infty} \left(\frac{e^n}{\pi}\right)^{1/n}$ $(\infty^0$ form)

$\therefore \log A = \lim_{n\to\infty} \frac{1}{n} \log\left(\frac{e^n}{\pi}\right) = \lim_{n\to\infty} \frac{n\log e - \log \pi}{n}$ $\left(\frac{\infty}{\infty} \text{ form}\right)$

$= \lim_{n\to\infty} \frac{\log e - 0}{1}$ {applying L'Hôpital's rule}

$\log A = 1 \Rightarrow A = e^1$ or $\lim_{n\to\infty} \left(\frac{e^n}{\pi}\right)^{1/n} = e$

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Illustration 59

Let $y=\sin^{-1}\left(\frac{3x^4+6x^2-1}{(x^2+1)^3}\right)$, then $\frac{dy}{dx}$ at $x^2=\tan 22\frac{1}{2}^0$ is

(A) $2-\sqrt{2}$ (B) $3(2-\sqrt{2})$ (C) $3(2+\sqrt{2})$ (D) $3\sqrt{2\sqrt{2}-2}$

Ans. (D)

Solution:

$$y=\sin^{-1}\left(\frac{3x^4+6x^2-1}{(x^2+1)^3}\right)=\sin^{-1}\left(\frac{3(x^2+1)^2-4}{(x^2+1)^3}\right)=\sin^{-1}\left(\frac{3}{x^2+1}-\frac{4}{(x^2+1)^3}\right)$$

Let $\frac{1}{x^2+1}=\sin\theta$

$y=\sin^{-1}(3\sin\theta-4\sin^3\theta)$

$$=\sin^{-1}(\sin 3\theta)=\begin{cases}3\theta; & 3\theta\in\left[0,\frac{\pi}{2}\right]\Rightarrow\theta\in\left[0,\frac{\pi}{6}\right]\\ \pi-3\theta; & 3\theta\in\left[\frac{\pi}{2},\frac{3\pi}{2}\right]\Rightarrow\theta\in\left[\frac{\pi}{6},\frac{\pi}{2}\right]\end{cases}$$

at $x^2=\tan 22\frac{1}{2}^0=\sqrt{2}-1$

$$\sin\theta=\frac{1}{\sqrt{2}-1+1}=\frac{1}{\sqrt{2}}\Rightarrow\theta=\frac{\pi}{4}$$

$$\therefore y=\pi-3\theta=\pi-3\sin^{-1}\left(\frac{1}{x^2+1}\right)$$

$$\frac{dy}{dx}=-3\frac{1}{\sqrt{1-\left(\frac{1}{x^2+1}\right)^2}}\left(-\frac{1}{x^2+1}\right).2x=\frac{6x}{(x^2+1)\sqrt{(x^2+1)^2-1}}$$

$$\frac{6.\sqrt{\sqrt{2}-1}}{(\sqrt{2}-1+1).\sqrt{2-1}}=\frac{6}{2}.\sqrt{2}\sqrt{\sqrt{2}-1}=3.\sqrt{2\sqrt{2}-1}$$

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Illustration 60

If $f'(5)=7$ then $\lim_{t\to0}\frac{f(5+t)-f(5-t)}{2t}=$

(A) 0 (B) 3.5 (C) 7 (D) 14

Ans. (C)

Solution:

$$f'(5)=7\ \lim_{t\to0}\frac{f(5+t)-f(5-t)}{2t}$$

$$\Rightarrow\lim_{t\to0}\left[\frac{f(5+t)-f(5)}{2t}+\frac{f(5-t)-f(5)}{-2t}\right]=\frac{f'(5)}{2}+\frac{f'(5)}{2}=f'(5)=7$$

Illustration 61

If $f(x)$ be a polynomial in x. Then the second derivative of $f(e^x)$, is

(A) $f''(e^x).e^x+f'(e^x)$ (B) $f''(e^x).e^{2x}+f'(e^x).e^{2x}$

(C) $f''(e^x)e^{2x}$ (D) $f''(e^x).e^{2x}+f'(e^x).e^x$

Ans. (D)

Solution:

$g=f(e^x)\Rightarrow g'=f'(e^x)e^x$

$\Rightarrow g=f(e^x)e^{2x}+e^x f'(e^x)$

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Illustration 62

$f'(x) = g(x)$ and $g'(x) = -f(x)$ for all real x and $f(5) = 2 = f'(5)$ then $f^2(10) + g^2(10)$ is

(A) 2 (B) 4 (C) 8 (D) None of these

Ans. (C)

Solution:

$f'(x) = g(x)$ and $g'(x) = -f(x)$

Now $\frac{d}{dx}[f^2(x) + g^2(x)] = 2f(x)f'(x) + 2g(x)g'(x) = 2f(x)g(x) - 2g(x)f(x) = 0$

$\therefore$ $f^2(x) + g^2(x)$ = constant

$f^2(5) + g^2(5) = 4 + 4 = 8$

$\therefore$ $f^2(10) + g^2(10) = 8$

Illustration 63

Let $f(x) = x\sin x\ \forall\ x \in \left(0, \frac{\pi}{2}\right)$ & g be the inverse of f in $\left(0, \frac{\pi}{2}\right)$, then $\left(\sqrt{3}\pi + 6\right)g'\left(\frac{\pi}{12}\right)$ is equal to

(A) 6 (B) 12 (C) 4 (D) 3

Ans. (B)

Solution:

$g'(f(x)) = \frac{1}{f'(x)}$

$f(x) = x\sin x$

When $f(x) = \frac{\pi}{12}$ $\Rightarrow g'\left(\frac{\pi}{12}\right) = \frac{1}{f'\left(\frac{\pi}{6}\right)}$ ...(i)

Now $f(x) = x\sin x$

$f'(x) = x\cos x + \sin x$

$\Rightarrow f'\left(\frac{\pi}{6}\right) = \frac{\pi}{6} \times \frac{\sqrt{3}}{2} + \frac{1}{2} = \frac{\sqrt{3}\pi + 6}{12}$ $\Rightarrow$ from (i)

$g'\left(\frac{\pi}{12}\right) = \frac{12}{\sqrt{3}} \Rightarrow \left(\sqrt{3}\pi + 6\right)g'\left(\frac{\pi}{12}\right) = 12$

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